

Chinmay Bansal

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EDUCATION

- **Purdue University** West Lafayette, IN
B.S. in Computer Science *Graduated December 2024*

PROFESSIONAL EXPERIENCE

- **SmarterD** San Francisco, California
Machine Learning Engineer Intern *May 2023 - October 2023*
 - Built an NLP-powered classification pipeline to automatically label 10,000+ enterprise vulnerability records by type, achieving 90% accuracy using a Naive Bayes classifier, enabling clients to identify high-risk threats instantly
 - Improved labeling by 65% using NLP and rule-based filtering to identify 500+ unique software names from 100,000+ vulnerability tickets, allowing clients to prioritize remediation based on software specific risk exposure
 - Implemented zero-shot classification using Meta BART to assign mitigation actions to 4,000+ unstructured solution descriptions with 92% accuracy, cutting incident response times by 50% through intelligent automation
- **Purdue University Computer Science** West Lafayette, IN
Teaching Assistant (CS 24000 - Programming in C) *August 2022 - December 2024*
 - Facilitated learning for 20+ students in weekly C programming labs by reinforcing theoretical concepts and providing hands-on guidance on practical programming techniques

PROJECTS

- **StockTeller** (stockteller.chinmaybansal.com)
 - Launched StockTeller, a full-stack AI-driven investment insights platform that parses 15,000+ Reddit posts and 1,000+ news articles across 6,500+ stock symbols using VADER and FinBERT sentiment models, surfacing emerging retail vs. institutional sentiment trends in real time to assist retail investors in idea discovery
 - Designed a multi-factor scoring engine that fused retail sentiment, technical indicators, and Alpha Vantage data to generate Buy/Sell/Hold ratings, increasing decision confidence and reducing research time from hours to seconds
 - Deployed the platform on Render using FastAPI, PostgreSQL, and React with intelligent dual-source caching, achieving sub-second responses for 65% of requests and improving first-load performance by 97% through progressive data loading and lazy stock validation
- **Spotifight**
 - Led the development of a real-time fantasy football full stack web app with over 700+ musical artists, leveraging React, Docker on AWS, MongoDB, integrated with Spotify and LastFM APIs
 - Architected 10+ RESTful API endpoints for backend-frontend communication, thoroughly tested with Postman; incorporated UI elements and custom error handling to enhance user experience
- **Wireminnow**
 - Engineered a packet analysis tool in C to parse PCAP files, inspect network traffic, and display packet details across 4+ protocols including IP, TCP, UDP, DHCP, and ICMP
 - Implemented features to decode timestamps, convert raw data into human-readable formats, and manage headers/payloads; tested on 5,000+ packets in simulated traffic
- **Xinu OS Development**
 - Developed essential OS functionality in C, including priority-based process scheduling, inter-process communication, and semaphore-driven synchronization for efficient multitasking
 - Constructed a thread-safe memory allocator utilizing a first-fit allocation strategy, block coalescing to minimize fragmentation, and integrity checks to enhance kernel reliability

TECHNICAL SKILLS AND RELEVANT COURSEWORK

- **Programming Languages:** Java, Python, C, C++, JavaScript, Swift
- **Tools and Skills:** MySQL, React.js, Next.js, MongoDB, Django, FastAPI, AWS, Git, NLP
- **Relevant Topics:** Operating Systems, Computer Networks, Systems Programming, Data Structures and Algorithms, Computer Security, Computer Architecture, Information Systems, Cryptography, Object-Oriented Programming, Software Engineering